

Team Worksheet — Lab 14

Baseline Capability — Cp, Cpk, Pp, Ppk and Non-Normal Data

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in lab-14-baseline-capability.xlsx.

3. What we produced

Deliverable: A baseline capability study for your capstone Y with a documented normality decision.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap capability in full: $C_p = (USL - LSL) / 6 \text{ sigma}$ measures spread only; C_{pk} also accounts for centring and is always $\leq C_p$.

Step 2. Recap the acceptance benchmarks: $Cpk \geq 1.33$ capable, ≥ 1.67 for critical characteristics, and the relationship between Cpk and sigma level.

Step 3. Learn the distinction most practitioners get wrong: Cp/Cpk use WITHIN-subgroup (short-term) variation; Pp/Ppk use TOTAL (long-term) variation.

Step 4. Understand the consequence: reporting Cpk on an unstable process flatters it, because between-subgroup drift is excluded from the estimate.

Step 5. Establish the rule — the process must be STABLE before capability means anything. Check stability with a control chart first.

Step 6. Test your capstone Y for normality using Anderson-Darling or Shapiro-Wilk. Read the p-value: $p < 0.05$ rejects normality.

Step 7. Recognise that cycle time, waiting time and defect counts are routinely non-normal and usually right-skewed — this is expected, not an error.

Step 8. For non-normal data, choose your route: apply a Box-Cox or Johnson transformation, or fit the correct distribution (Weibull, lognormal, exponential) directly.

Step 9. Apply the transformation, re-test normality, then compute capability on the transformed scale — and report results in the ORIGINAL units.

Step 10. Calculate both Cpk and Ppk for your capstone Y and explain the gap between them. A large gap means the process drifts between subgroups.

Step 11. Record the baseline capability in your capstone pack alongside the Lab 3 sigma level.

7. Self-check

You have confirmed stability before computing capability, tested and documented normality with a p-value, handled non-normality by transformation or distribution fitting, and reported both Cpk and Ppk with an explanation of the gap.

Complete? ☐ Yes ☐ Not yet

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